Performance modeling for Computer systems
(Endsem 40 Marks, 90 mins)

1. Let $A(t)$ and $D(t)$ denote two independent Poisson processes with rates λ and μ respectively.Condsider 3 cases $\lambda > \mu, \lambda < \mu$ and $\lambda = \mu$ and for each case analyze the following. Consider $X(t) = A(t) - D(t)$. Is $X(t)$ a renewal process ? Why ? Show that $X(t)$ is a a CTMC and obtain its Q matrix. Also obtain its stationary dsitribution, if it exists. (7mks) What can you intuitively (no proofs needed) comment about $N(t) = max(X(t), 0)$? How does its Q matrix look like? (3mks)

2. Consider and $M/G/1/1$ system with two streams of arrivals, named Stream 1 and Stream 2. Stream i jobs arrive according to a Poisson Process with rate λ_i and have a service time distribution denoted by G_i for $i = 1, 2$. Note that there is only one server and no waiting position. Using Renewal reward theorems, obtain $\bar{B}_i$ which is the time average occupancy of the server by stream i for $i = 1, 2$.

3. For the $M/G/1$ queue, derive the expression for the mean number of jobs in the system using little's law, busy period analysis and renewal reward theorem if needed.

4. Consider a positive recurrent aperiodic CTMC with finite state space. Show using renewal reward theorem that the stationary distribution satisfies $\pi_i = \frac{1}{|q_{ii}|\mu_{ii}}$ where μ_{ii} is the mean recurrence time for state i.